

Name: _____

This practice exam covers Chapters 9 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. An unknown compound has a molar mass of 48 g/mol. A sample is vaporized at 124. °C. At a pressure of 0.97 atm, the gas occupies 957 mL. Determine the mass of the sample. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

2. The reaction $2\text{NaHCO}_3(\text{s}) \longrightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 20.8 L. After 18.5 g of NaHCO_3 completely reacts while all other reactants are present in excess at 21. °C, determine the pressure of H_2O . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

3. A gas mixture has a total pressure of 0.9305 atm. The mole fraction of O_2 is 0.350. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

4. A gas sample is trapped inside a syringe with its tip sealed. Initially the gas has a pressure of 261 kPa and occupies 11.0 L. The pressure is then increased to 770. kPa while the temperature remains constant. Determine the final volume. Justify your answer.

ANSWER: _____

5. A 8.61-L tank at 81.6 °C and 1.30 atm contains 9.86 g of hydrogen sulfide (H_2S) and 9.8 g of dinitrogen monoxide (N_2O). Determine the mole fraction of dinitrogen monoxide in the mixture. Justify your answer.

ANSWER: _____

6. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 251 MPa. The cover of a circular observation port can safely withstand a maximum force of 4.5 MN. Determine the largest allowable diameter of the observation port. Report your answer in SI units and justify your answer.

ANSWER: _____

7. A sample of 23 g of hydrogen peroxide decomposes completely to produce water and oxygen gas in a sealed container. The container initially contains 0.318 mol of argon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of argon after the reaction is complete.

ANSWER: _____

8. A gas is confined beneath a frictionless piston with a circular cross section of diameter 11-cm. A steady downward force of 51-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

9. The reaction $\text{Fe(s)} + 2\text{HCl(aq)} \longrightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ is carried out in a sealed, initially evacuated vessel with a volume of 24.2 L. After 5.1 g of HCl completely reacts while all other reactants are present in excess at 14.°C, determine the pressure of H_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

10. A sealed flask is evacuated and then filled with an unknown gas. The sample has a mass of 2.490 g, occupies 0.750 L at 100.0 kPa and 293.15 K. Possible identities are: F_2 , H_2O , CH_4 , NO , Ar , CO_2 , UF_6 , SO_2 , CO , H_2S , C_2H_2 , Kr , Ne , He , SF_6 , Cl_2 , H_2 , HBr , Xe . Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

11. The vapor pressure of a volatile solvent is measured at 38°C. The sample is then heated to 60°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

12. O_2 is collected over water. The total pressure of the gas sample is 769.0 torr, and the vapor pressure of water is 27 torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

13. The reaction $\text{CaC}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow \text{C}_2\text{H}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{aq})$ is carried out in a sealed, initially evacuated vessel with a volume of 12.6 L. After 6.98 g of CaC_2 completely reacts while all other reactants are present in excess at 8.14°C , determine the pressure of C_2H_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

14. Consider the reaction $\text{N}_2\text{O}_4(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$. If 6.5 cm^3 of N_2O_4 gas is completely consumed at the same temperature and pressure, what volume of NO_2 gas is produced? Justify your answer.

ANSWER: _____

15. Lauren is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 78.2-cm. After filling, the pressure inside the tank is 131 atm and the temperature is 300. K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

16. A gas sample is trapped inside a syringe with its tip sealed. Its initial volume is 7.2 L. After the piston is moved, the gas occupies 1.7 L and its pressure is measured to be 476 kPa. The temperature remains constant throughout. Determine the initial pressure. Justify your answer.

ANSWER: _____

17. A 585-mL sample of dichloromethane (CH_2Cl_2) is initially at 497 K. Its boiling point is $40.^\circ\text{C}$ (313.15 K). The pressure is then reduced to 62% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

18. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 446 MPa. The cover of a circular inspection hatch can safely withstand a maximum force of 4.1 MN. Determine the largest allowable diameter of the inspection hatch. Report your answer in SI units and justify your answer.

ANSWER: _____

19. A 4.7 g sample of propylene gas (C_3H_6) is held in a gas collection bulb at 14.1°C and 1.08 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

20. A liquid is boiling at 77°C in a chamber where the external pressure is 170 kPa. It occupies a volume of 4 L. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

21. The reaction $2\text{CO}(\text{g}) + 4\text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_5\text{OH}(\text{g}) + \text{H}_2\text{O}(\text{g})$ occurs at constant temperature and pressure. If 3.59 cm^3 of $\text{C}_2\text{H}_5\text{OH}$ gas reacts completely, what volume of CO gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

22. A laboratory apparatus contains gas beneath a circular movable piston of diameter 12.9-cm. The gas column is 13-cm high before any compression. A constant downward force of 74.9-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

23. An unknown gas has a density of $1.4 \times 10^3 \frac{\text{kg}}{\text{m}^3}$ at 93.9 kPa and $26.^\circ\text{C}$. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

24. Hydrogen gas is collected over water. The total pressure is 739 torr, and the vapor pressure of water is 34.5 torr. Determine the partial pressure of the hydrogen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

25. The reaction $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 5.12 mL of nitrogen dioxide reacts completely, what volume of nitric oxide is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

26. A 7.9 g sample of neon gas (Ne) is contained in a laboratory flask at 39.°C and 1.1 atm. Determine the volume occupied assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

27. A rigid 13-L tank at 81.1°C contains only dinitrogen monoxide (N_2O) and carbon dioxide (CO_2). Assume both gases behave ideally. The tank contains 7.1 g of dinitrogen monoxide. The total amount of gas is 0.32 mol. Determine the pressure contributed by the second gas. Justify your answer.

ANSWER: _____

28. A gas is contained beneath a square piston with sides of 11.2-cm. The gas pressure is measured to be 11.7 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

29. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs at constant temperature and pressure. If 5.05 cm³ of Cl_2 gas reacts completely, what volume of H_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

30. A laboratory apparatus contains gas beneath a circular movable piston of diameter 7.9-cm. The gas column is 9.55-cm high before any compression. A constant downward force of 78.8-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

31. The reaction $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ occurs at constant temperature and pressure. If 7.92 cm^3 of C_2H_6 gas reacts completely, what volume of O_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

32. An unknown compound has a molar mass of 38 g/mol. A sample is vaporized at 100°C . At a pressure of 1.25 atm, the gas occupies 646 mL. Determine the mass of the sample. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

33. The reaction $2\text{CO} + 4\text{H}_2 \longrightarrow \text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$ occurs with all gaseous substances at the same temperature and pressure. If 3.50 mL of H_2 is completely consumed, what volume of $\text{C}_2\text{H}_5\text{OH}$ is consumed or produced, as appropriate? Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

34. A gas mixture contains only He and CO_2 . Their partial pressures are 0.412 atm and 0.2331 atm, respectively. Determine the total pressure of the mixture. Report your answer in SI units. Justify your answer.

ANSWER: _____

35. A fixed amount of gas changes from 20°C to 0°C while its pressure changes from 9.01 atm to 6.22 atm. Determine the final volume as a percentage of the initial volume. Justify your answer.

ANSWER: _____

36. A sample of NO is transferred into a rigid reaction vessel having a measured volume of 4.13 L. After the gas reaches 18.°C, the pressure inside is measured to be 0.745 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

37. A laboratory apparatus contains gas beneath a circular movable piston of diameter 5.30-cm. The gas column is 8.43-cm high before any compression. A constant downward force of 40.2-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

38. H₂ is collected over water at a temperature where the vapor pressure of water is 17 torr. The dry gas has a partial pressure of 754 torr. Determine the total pressure of the collected gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

39. The partial pressure of N₂ in a gas mixture is 0.554 atm. The mole fraction of N₂ is 0.350. Determine the total pressure of the mixture. Report your answer in SI units. Justify your answer.

ANSWER: _____

40. John is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 62.8-cm. After filling, the pressure inside the tank is 201 atm and the temperature is 292 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

41. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 731.6 torr, and the mercury is 47.1 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

42. The reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 9.9cm^3 of oxygen is consumed, what volume of water vapor is produced or consumed? Justify your answer.

ANSWER: _____

43. An unknown compound has a molar mass of 28g/mol . A sample is vaporized at $77.^{\circ}\text{C}$. At a pressure of 1.5atm , the gas occupies 970mL . Determine the mass of the sample. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

44. A sample of Ar is transferred into a volumetric gas flask having a measured volume of 3.8L . After the gas reaches 3.3°C , the pressure inside is measured to be 0.329atm . Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

45. A gas sample is trapped inside a syringe with its tip sealed. Initially the gas has a pressure of $2.0 \times 10^2\text{kPa}$ and occupies 15L . The pressure is then increased to $540.\text{kPa}$ while the temperature remains constant. Determine the final volume. Justify your answer.

ANSWER: _____

46. Consider the reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$. If 3.14cm^3 of Cl_2 gas is completely consumed at the same temperature and pressure, what volume of HCl gas is produced? Justify your answer.

ANSWER: _____

47. A 482-mL sample of pentane (C_5H_{12}) is initially at 503K . Its boiling point is $36.^{\circ}\text{C}$ (309.15K). The pressure is then reduced to 84% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^{\circ}\text{C}$. Justify your answer.

ANSWER: _____

48. The vapor pressure of a volatile solvent is measured at 92°C. The sample is then cooled to 54°C while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

49. A liquid is boiling at 47°C in a chamber where the external pressure is 57.9 kPa. It occupies a volume of 3 L. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

50. A 13.0 g sample of ethane gas (C_2H_6) is stored in a rigid cylinder at 21.1°C and 0.922 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

51. A gas is contained beneath a square piston with sides of 9.8-cm. The gas pressure is measured to be 20. kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

52. Consider the reaction $2\text{C}_4\text{H}_{10}(\text{g}) + 13\text{O}_2(\text{g}) \longrightarrow 8\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{g})$. If 13. cm^3 of C_4H_{10} gas is completely consumed at the same temperature and pressure, what volume of CO_2 gas is produced? Justify your answer.

ANSWER: _____

53. The reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 5.69 cm^3 of sulfur dioxide is consumed, what volume of sulfur trioxide is produced or consumed? Justify your answer.

ANSWER: _____

54. The reaction $\text{Cl}_2 + \text{H}_2 \longrightarrow 2\text{HCl}$ occurs with all gaseous substances at the same temperature and pressure. During the reaction, 5.2 mL of HCl reacts completely. Determine the corresponding volume of Cl_2 . Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

55. A fixed amount of gas is initially at 12.1°C and 4.71 atm. It is later found to have a volume that is 49% smaller than its initial volume and a pressure of 7.00 atm. Determine the final temperature. Report your answer in SI units and justify your answer.

ANSWER: _____

56. The reaction $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 10.3 L. After 4.32 g of H_2O_2 completely reacts while all other reactants are present in excess at 5.0°C, determine the pressure of O_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

57. A rigid cylinder containing an ideal gas is heated while no gas enters or leaves the cylinder. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

58. A sample reported to be H_2S has a mass of 7.2 g and occupies 5.41 L at 93.5 kPa and 16.°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

59. A sample of 8.7 g of ammonium chloride reacts completely with excess sodium hydroxide to produce sodium chloride, water, and ammonia gas in a sealed container. The container initially contains 0.32 mol of helium gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of helium after the reaction is complete.

ANSWER: _____

60. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 420. MPa. The window of a circular viewport can safely withstand a maximum force of 5.28 MN. Determine the largest allowable diameter of the viewport. Report your answer in SI units and justify your answer.

ANSWER: _____

61. A gas is reported to be C_2H_6 . Its density is measured as $2.3 \times 10^3 \frac{\text{g}}{\text{m}^3}$ at 120 kPa and 20.7°C . Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

62. A sample reported to be He has a mass of 160 g and occupies 11.9 L at 96 kPa and $21.^\circ\text{C}$. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

63. A gas mixture has a total pressure of 1.031 atm. The mole fraction of O_2 is 0.780. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

64. A 7.8 g sample of chlorine gas (Cl_2) is held in a gas collection bulb at 22.4°C and 1.3 atm. Determine the volume occupied assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

65. Ryan is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 29.2-cm and an internal height of 126-cm. After filling, the pressure inside the tank is 205 atm and the temperature is 325 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

66. An unknown compound is completely vaporized at $116.^\circ\text{C}$. At a pressure of 1.2 atm, the gas occupies 940 mL and has a mass of 1.47 g. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

67. Consider the reaction $2\text{C}_4\text{H}_{10}(\text{g}) + 13\text{O}_2(\text{g}) \longrightarrow 8\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{g})$. If 3.3cm^3 of C_4H_{10} gas is completely consumed at the same temperature and pressure, what volume of CO_2 gas is produced? Justify your answer.

ANSWER: _____

68. The reaction $2\text{HBr}(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g}) + \text{Br}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 6.8 mL of bromine vapor reacts completely, what volume of chlorine gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

69. A gas is reported to be N_2O . Its density is measured as $1.4 \times 10^3 \frac{\text{g}}{\text{m}^3}$ at 120 kPa and 22.9°C . Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

70. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 558 MPa. The piston of a circular piston can safely withstand a maximum force of 5.18 MN. Determine the largest allowable diameter of the circular piston. Report your answer in SI units and justify your answer.

ANSWER: _____

71. A gas mixture contains 8.5 g of bromine pentafluoride (BrF_5) and 8.3 g of dinitrogen monoxide (N_2O). Determine the mole fraction of bromine pentafluoride in the mixture. Justify your answer.

ANSWER: _____

72. The vapor pressure of a volatile solvent is measured at 31°C . The sample is then heated to 47°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

73. A liquid is boiling at 38°C in a chamber where the external pressure is 47.2 kPa . It occupies a volume of 4 L . Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

74. Consider the reaction $2\text{SO}_3(\text{g}) \longrightarrow 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$. If 6.77 cm^3 of SO_3 gas is completely consumed at the same temperature and pressure, what volume of O_2 gas is produced? Justify your answer.

ANSWER: _____

75. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 723.8 torr , and the mercury is 150 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

76. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 13.7 mL of hydrogen chloride reacts completely, what volume of hydrogen gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

77. A compressed-gas cylinder is connected to a large evacuated storage bag. The cylinder has an internal diameter of 4.8-cm , an internal height of 24-cm , and initially contains gas at 5.72 atm . The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

78. The reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{HBr}(\text{g})$ occurs at constant temperature and pressure. If 5.87 cm^3 of HBr gas reacts completely, what volume of Br_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

79. A gas is contained beneath a square piston with sides of 5.9-cm. The gas pressure is measured to be 7.9kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

80. A fixed amount of gas changes from 25.1°C to -17°C while its pressure changes from 5.6atm to 16atm. Determine the final volume as a percentage of the initial volume. Justify your answer.

ANSWER: _____

81. A gas is contained beneath a square piston with sides of 10.-cm. The gas pressure is measured to be 14.6kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

82. A gas is confined beneath a frictionless piston with a circular cross section of diameter 11.4-cm. A steady downward force of 78-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

83. A gas sample is trapped inside a syringe with its tip sealed. Initially the gas has a pressure of 240kPa and occupies 10.5L. The pressure is then increased to 740kPa while the temperature remains constant. Determine the final volume. Justify your answer.

ANSWER: _____

84. The reaction $\text{CO(g)} + \text{H}_2\text{O(g)} \longrightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$ occurs at constant temperature and pressure. If 4.7 cm³ of H₂O gas reacts completely, what volume of CO₂ gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

85. An unknown gas has a mass of 28 g and occupies 5.54 L at 140 kPa and 18.8°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

86. Thomas is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 33.8-cm and an internal height of 120.-cm. After filling, the pressure inside the tank is 148 atm and the temperature is 300. K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

87. A chemist isolates an unknown gas in a sealed vessel. The sample has a mass of 2.683 g, occupies 0.500 L at 101.3 kPa and 298.15 K. Possible identities are: Cl_2 , H_2O , CO_2 , UF_6 , C_2H_2 , H_2 , CO , HBr , SO_2 , F_2 , Xe , Kr , H_2S , NO , He , SF_6 , Ne , Ar , CH_4 . Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

88. A sample reported to be Ne has a mass of 6.82 g and occupies 3.31 L at 146 kPa and 17.1°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

89. H_2 is collected over water. The total pressure of the gas sample is 745 torr, and the vapor pressure of water is 24 torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

90. A sample of O_2 is collected over water. The partial pressure of the dry gas is 736.4 torr, and the vapor pressure of water is 32.2 torr. Determine the total pressure of the gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

91. A sample reported to be HF has a mass of 6.6g and occupies 4.83L at 163kPa and 16.°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

92. A closed-end mercury manometer connected to a gas sample has a height difference of 46mmHg. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

93. An unknown compound is completely vaporized at 80.1°C. At a pressure of 1.1 atm, the gas has a density of 1.9g/L. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

94. A sample reported to be C₃H₈ has a mass of 18g and occupies 3.3L at 152kPa and 9.8°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

95. A chemist isolates an unknown gas in a sealed vessel. The sample has a mass of 2.161 g, occupies 1.250 L at 150.0 kPa and 313.15 K. Possible identities are: He, SF₆, NO, H₂, Kr, Cl₂, Ne, CO₂, C₂H₂, F₂, Ar, H₂O, CO, Xe, UF₆, SO₂, HBr, H₂S, CH₄. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

96. An unknown compound is completely vaporized at 110°C. At a pressure of 1.18 atm, the gas has a density of 1.4g/L. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

97. The reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 13.7 mL of hydrogen gas reacts completely, what volume of water vapor is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

98. A 7.5 g sample of ethane gas (C_2H_6) is contained in a laboratory flask at 23.6°C and 1.34 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

99. A compressed-gas cylinder is connected to a large evacuated storage bag. The cylinder has an internal diameter of 5.00-cm, an internal height of 29-cm, and initially contains gas at 3.2 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

100. A gas sample is trapped inside a syringe with its tip sealed. After the piston is moved, the gas has a pressure of 883 kPa and occupies 1.0 L. Its initial pressure was 240 kPa, and the temperature remained constant throughout. Determine the initial volume. Justify your answer.

ANSWER: _____

101. The reaction $2\text{NH}_4\text{Cl}(\text{s}) + \text{Ca}(\text{OH})_2(\text{s}) \longrightarrow \text{CaCl}_2(\text{s}) + 2\text{NH}_3(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ is carried out in a sealed, initially evacuated vessel with a volume of 13 L. After 3.54 g of NH_4Cl completely reacts while all other reactants are present in excess at 30.5°C , determine the pressure of NH_3 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

102. The reaction $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ occurs at constant temperature and pressure. If 4.0 cm^3 of CH_4 gas reacts completely, what volume of CO_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

103. A liquid is boiling at 34°C in a chamber where the external pressure is 27.6 kPa . It occupies a volume of 3 L . Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

104. A rigid 13.9-L tank at 86.2°C contains only ethane (C_2H_6) and dinitrogen difluoride (N_2F_2). Assume both gases behave ideally. The tank contains $10.\text{ g}$ of ethane and 8.19 g of dinitrogen difluoride. Determine the decimal fraction of the total pressure that would remain if all of the dinitrogen difluoride were removed while the temperature and volume remained unchanged. Justify your answer.

ANSWER: _____

105. A rigid 8.71-L tank at $101.^{\circ}\text{C}$ contains only propylene (C_3H_6) and ethane (C_2H_6). Assume both gases behave ideally. The tank contains 9.4 g of propylene and 6.63 g of ethane. Determine the ratio of the pressure produced by propylene to that produced by ethane. Justify your answer.

ANSWER: _____

106. A sample of an unknown gas is transferred into a dry gas bulb. The sample has a mass of 0.541 g , occupies 0.500 L at 101.3 kPa and 293.15 K . Possible identities are: Kr, SO_2 , CO, UF_6 , HBr, C_2H_2 , H_2 , CO_2 , NO, Ne, Cl_2 , H_2S , H_2O , F_2 , He, Ar, SF_6 , CH_4 , Xe. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

107. A gas mixture contains only N_2 and CO_2 . Their partial pressures are 0.2457 atm and 0.2963 atm , respectively. Determine the total pressure of the mixture. Report your answer in SI units. Justify your answer.

ANSWER: _____

108. A sample of an unknown gas is transferred into a dry gas bulb. The sample has a mass of 0.081 g , occupies 1.000 L at 100.0 kPa and 298.15 K . Possible identities are: Xe, H_2O , UF_6 , SF_6 , CO_2 , NO, H_2 , CO, SO_2 , Ne, C_2H_2 , F_2 , HBr, Ar, CH_4 , Cl_2 , H_2S , Kr, He. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

109. A sample of 14 g of potassium bicarbonate reacts completely with excess hydrochloric acid to produce potassium chloride, water, and carbon dioxide gas in a sealed container. The container initially contains 0.43 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: _____

110. A rigid cylinder containing an ideal gas is heated while no gas enters or leaves the cylinder. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

111. A gas mixture contains 12 g of dinitrogen monoxide (N_2O) and 7.5 g of xenon (Xe). Determine the mole fraction of xenon in the mixture. Justify your answer.

ANSWER: _____

112. A gas is contained beneath a square piston with sides of 7.53-cm. The gas pressure is measured to be 7.30 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

113. An unknown gas has a density of $540 \frac{\text{kg}}{\text{m}^3}$ at 101 kPa and 12.2°C . Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

114. A 29.3 g sample of neon gas (Ne) is inside a lecture bottle at $33.^\circ\text{C}$ and 1.5 atm. Calculate the volume of the sample assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

115. The reaction $2\text{HCl}(\text{g}) + \text{NH}_3(\text{g}) \longrightarrow \text{NH}_4\text{Cl}(\text{s})$ occurs with all gases measured at the same temperature and pressure. If 9.92 cm^3 of hydrogen chloride is consumed, what volume of ammonia is produced or consumed? Justify your answer.

ANSWER: _____

116. A research balloon contains 2.6L of helium at -11.3°C while indoors. It is then taken outside, where the air temperature is 23°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

117. The vapor pressure of a liquid is measured at 60°C . Additional liquid of the same substance is added while the temperature is held constant and liquid remains present before and after the addition. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

118. A flexible weather balloon contains 2.7L of argon at -11.8°C while indoors. It is then taken outside, where the air temperature is 21.5°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

119. Oxygen gas is collected over water. The measured pressure is 747.6torr, and the vapor pressure of water is 24torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

120. A liquid has a vapor pressure of 112kPa at 82°C . The external pressure is 1.00 atm. Determine whether the liquid will boil. Report the vapor pressure and justify your answer.

ANSWER: _____

121. A rigid container is evacuated and then refilled to the same temperature and pressure with a different ideal gas whose molar mass is smaller. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

122. A 833-mL sample of ethane (C_2H_6) is initially at 343 K. Its boiling point is $-89.^\circ\text{C}$ (184.15 K). The pressure is then reduced to 37% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

123. A rigid 14.7-L tank at 130°C contains only dinitrogen monoxide (N_2O) and sulfur tetrafluoride (SF_4). Assume both gases behave ideally. The total pressure is 1.2 atm, and the mole ratio of dinitrogen monoxide:sulfur tetrafluoride is 2.48 : 1. Determine the pressure produced by the second gas. Justify your answer.

ANSWER: _____

124. A flexible gas bag contains 2.51 L of nitrogen at 26.0°C while indoors. It is then taken outside, where the air temperature is -8.9°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

125. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs at constant temperature and pressure. If 5.55 cm^3 of Cl_2 gas reacts completely, what volume of HCl gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

126. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 735.6 torr and the mercury is 130. mmHg higher on the end open to the atmosphere. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

127. A high-pressure gas cylinder is allowed to fill a large empty collection bag. The cylinder has an internal diameter of 7.7-cm, an internal height of 28-cm, and initially contains gas at 4.49 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

128. A 502-mL sample of boron trichloride (BCl_3) is initially at 324 K. Its boiling point is $12.^\circ\text{C}$ (285.15 K). The pressure is then reduced to 59% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

129. A sample of an ideal gas in a cylinder with a movable piston is compressed slowly while its temperature is kept constant. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

130. N_2 is collected over water at a temperature where the vapor pressure of water is 26 torr. The dry gas has a partial pressure of 758 torr. Determine the total pressure of the collected gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

131. An unknown gas has a density of $1.3 \times 10^3 \frac{\text{kg}}{\text{m}^3}$ at 104 kPa and $22.^\circ\text{C}$. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

132. An unknown compound has a molar mass of 42 g/mol. A sample is vaporized at $131.^\circ\text{C}$. At a pressure of 1.2 atm, the gas occupies 920 mL. Determine the mass of the sample. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

133. The reaction $\text{N}_2\text{O}_4(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 3.23 cm^3 of dinitrogen tetroxide is consumed, what volume of nitrogen dioxide is produced or consumed? Justify your answer.

ANSWER: _____

134. The reaction $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$ occurs with all gaseous substances at the same temperature and pressure. If the reaction consumes 3.32 mL of CaO , determine the volume of CO_2 . Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

135. A 464-mL sample of dimethyl ether ($(\text{CH}_3)_2\text{O}$) is initially at 376 K. Its boiling point is $-25.^\circ\text{C}$ (248.15 K). The pressure is then increased to 2.5 times its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

136. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 3.39 mL of hydrogen gas reacts completely, what volume of hydrogen chloride is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

137. HF is collected in a volumetric gas flask having a measured volume of 6.00 L. After the gas reaches 9.4°C , the pressure inside is measured to be 0.66 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

138. Additional ideal gas is introduced into a rigid container while the temperature is held constant. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

139. A 11.8-L tank at 130°C and 1.19 atm contains 0.22 mol of dinitrogen tetroxide (N_2O_4) and 0.303 mol of nitrogen trifluoride (NF_3). Determine the mole fraction of nitrogen trifluoride in the mixture. Justify your answer.

ANSWER: _____

140. The reaction $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 7.7 mL of water vapor reacts completely, what volume of oxygen gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

141. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 4.3 cm^3 of hydrogen chloride is consumed, what volume of hydrogen is produced or consumed? Justify your answer.

ANSWER: _____

142. A flexible latex balloon contains 2.5 L of helium at -8.13°C while indoors. It is then taken outdoors, where the air temperature is 31.6°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

143. A fixed amount of gas is initially at 22.9°C and 4.8 atm. It is later found to have a volume that is 51% smaller than its initial volume and a pressure of 12 atm. Determine the final temperature. Report your answer in SI units and justify your answer.

ANSWER: _____

144. Ian is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 67.7-cm. After filling, the pressure inside the tank is 147 atm and the temperature is 294 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

145. A movable piston traps a fixed amount of gas inside a cylinder. Initially the gas has a pressure of 260 kPa and occupies 12 L. The pressure is then increased to 660 kPa while the temperature remains constant. Determine the final volume. Justify your answer.

ANSWER: _____

146. A fixed amount of gas is initially at 4.01 atm and 19.9°C. It is then brought to a new state in which its temperature is increased to 46°C and its volume is 62% larger than its initial volume. Determine the final pressure. Report your answer in SI units and justify your answer.

ANSWER: _____

147. The reaction $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$ occurs with all gaseous substances at the same temperature and pressure. If 5.34 mL of CaO is completely consumed, what volume of CO_2 is consumed or produced, as appropriate? Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

148. A liquid has a vapor pressure of 91.9 kPa at 31°C. The external pressure is 1.00 atm. Determine whether the liquid will boil. Report the vapor pressure and justify your answer.

ANSWER: _____

149. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 173 MPa. The cover of a circular inspection hatch can safely withstand a maximum force of 3.9 MN. Determine the largest allowable diameter of the inspection hatch. Report your answer in SI units and justify your answer.

ANSWER: _____

150. A 611-mL sample of silane (SiH_4) is initially at 351 K. Its boiling point is -112°C (161.15 K). The pressure is then increased to 1.2 times its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest °C. Justify your answer.

ANSWER: _____

151. A sample of Kr is transferred into a calibrated glass bulb having a measured volume of 5.6 L. After the gas reaches 10.3°C, the pressure inside is measured to be 0.70 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

152. An unknown gas has a density of $950 \frac{\text{kg}}{\text{m}^3}$ at 90. kPa and 5.23°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

153. A laboratory apparatus contains gas beneath a circular movable piston of diameter 5.7-cm. The gas column is 13-cm high before any compression. A constant downward force of 39.3-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

154. A gas is confined beneath a frictionless piston with a circular cross section of diameter 7.94-cm. A steady downward force of 55-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

155. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 772 torr, and the mercury is 53 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

156. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 733 torr, and the mercury is 41 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

157. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 757.1 torr, and the mercury is 63.6 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

158. The vapor pressure of a liquid is measured at 56°C. Additional liquid of the same substance is added while the temperature is held constant and liquid remains present before and after the addition. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

159. A compressed-gas cylinder is connected to a large evacuated storage bag. The cylinder has an internal diameter of 7.62-cm, an internal height of 29-cm, and initially contains gas at 6.2 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

160. A gas mixture contains O_2 , Ne, and N_2 . The total pressure is 1.02 atm. The mole fractions of O_2 and Ne are 0.150 and 0.150, respectively. Determine the partial pressure of N_2 . Report your answer in SI units. Justify your answer.

ANSWER: _____

161. A 407-mL sample of nitrogen (N_2) is initially at 157 K. Its boiling point is $-196.^{\circ}C$ (77.15 K). The pressure is then reduced to 20% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^{\circ}C$. Justify your answer.

ANSWER: _____

162. A gas mixture has a total pressure of 1.059 atm. The mole fraction of O_2 is 0.350. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

163. Charles is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 38.0-cm and an internal height of 123-cm. After filling, the pressure inside the tank is 186 atm and the temperature is 287 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

164. The vapor pressure of a liquid is measured at $38.^{\circ}C$. Additional liquid of the same substance is added while the temperature is held constant and liquid remains present before and after the addition. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

165. The vapor pressure of an unknown compound is measured at 36°C. Additional liquid of the same substance is added while the temperature is held constant and liquid remains present before and after the addition. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

166. Oxygen gas is collected over water. The measured pressure is 744.9 torr, and the vapor pressure of water is 19.5 torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

167. An unknown gas has a density of $773 \frac{\text{kg}}{\text{m}^3}$ at 98 kPa and 34.3°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

168. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 191 MPa. The window of a circular viewport can safely withstand a maximum force of 3.30 MN. Determine the largest allowable diameter of the viewport. Report your answer in SI units and justify your answer.

ANSWER: _____

169. A gas mixture contains 0.418 mol of hydrogen chloride (HCl) and 0.321 mol of dinitrogen difluoride (N_2F_2). Determine the mole fraction of hydrogen chloride in the mixture. Justify your answer.

ANSWER: _____

170. A sample of 22 g of sodium nitrite decomposes completely to produce sodium oxide, nitrogen monoxide, and oxygen gas in a sealed container. The container initially contains 0.47 mol of helium gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of helium after the reaction is complete.

ANSWER: _____

171. Allison is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 68.0-cm. After filling, the pressure inside the tank is 131 atm and the temperature is 326 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

172. CF_4 is sealed inside a calibrated glass bulb having a measured volume of 3.3 L. After the gas reaches 3.78°C , the pressure inside is measured to be 0.39 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

173. A flexible latex balloon contains 2.70 L of neon at 21°C while indoors. It is then taken outdoors, where the air temperature is -8.6°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

174. A sealed flask is evacuated and then filled with an unknown gas. The sample has a mass of 4.612 g, occupies 1.250 L at 100.0 kPa and 273.15 K. Possible identities are: UF_6 , C_2H_2 , Ne, CH_4 , Xe, Ar, NO, F_2 , Cl_2 , HBr, SF_6 , H_2 , H_2O , He, H_2S , CO_2 , SO_2 , CO, Kr. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

175. The partial pressure of O_2 in a gas mixture is 0.294 atm. The mole fraction of O_2 is 0.180. Determine the total pressure of the mixture. Report your answer in SI units. Justify your answer.

ANSWER: _____

176. A rigid container is evacuated and then refilled to the same temperature and pressure with a different ideal gas whose molar mass is larger. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

177. A flexible weather balloon contains 2.96 L of helium at -23°C while indoors. It is then taken outside, where the air temperature is 25.5°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

178. A gas cylinder is opened into a large empty flexible reservoir. The cylinder has an internal diameter of 8.90-cm, an internal height of 23-cm, and initially contains gas at 4.08 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

179. Consider the reaction $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$. If 6.34 cm^3 of NO gas is completely consumed at the same temperature and pressure, what volume of NO_2 gas is produced? Justify your answer.

ANSWER: _____

180. The reaction $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$ occurs with all gaseous substances at the same temperature and pressure. If 13.3 mL of O_2 is completely consumed, what volume of H_2O is consumed or produced, as appropriate? Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

181. Consider the reaction $2\text{NOCl}(\text{g}) \longrightarrow 2\text{NO}(\text{g}) + \text{Cl}_2(\text{g})$. If 4.77 cm^3 of NOCl gas is completely consumed at the same temperature and pressure, what volume of Cl_2 gas is produced? Justify your answer.

ANSWER: _____

182. A rigid reaction vessel is filled with SF_6 having a measured volume of 8.85 L. After the gas reaches 21.6°C , the pressure inside is measured to be 0.822 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

183. A pressurized gas cylinder is connected to a large evacuated balloon. The cylinder has an internal diameter of 5.4-cm, an internal height of 23-cm, and initially contains gas at 6.16 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

184. A rigid 7.27-L tank at 140.°C contains only chlorine (Cl_2) and dinitrogen difluoride (N_2F_2). Assume both gases behave ideally. The tank contains 6.67 g of chlorine and 14 g of dinitrogen difluoride. Determine the ratio of the pressure produced by chlorine to that produced by dinitrogen difluoride. Justify your answer.

ANSWER: _____

185. A gas mixture contains 7.7 g of ammonia (NH_3) and 11.5 g of oxygen (O_2). Determine the mole fraction of oxygen in the mixture. Justify your answer.

ANSWER: _____

186. The reaction $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 10.L. After 5.7 g of H_2O_2 completely reacts while all other reactants are present in excess at 28.°C, determine the pressure of O_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

187. An unknown gas has a mass of 17.1 g and occupies 14.0L at 180kPa and 11.1°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

188. The reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 11.7 cm³ of oxygen is consumed, what volume of sulfur dioxide is produced or consumed? Justify your answer.

ANSWER: _____

189. A sample of 18.7 g of barium carbonate reacts completely with excess hydrochloric acid to produce barium chloride, water, and carbon dioxide gas in a sealed container. The container initially contains 0.45 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: _____

190. A gas sample is trapped inside a syringe with its tip sealed. Its initial pressure is 236 kPa and its initial volume is 10.2 L. The piston is moved until the gas occupies 3.4 L while the temperature remains constant. Determine the final pressure. Justify your answer.

ANSWER: _____

191. The reaction $2\text{NO}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 3.30 cm^3 of nitric oxide is consumed, what volume of oxygen is produced or consumed? Justify your answer.

ANSWER: _____

192. A sample of H_2 is collected over water. The partial pressure of the dry gas is 682.9 torr, and the vapor pressure of water is 31 torr. Determine the total pressure of the gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

193. A rigid cylinder containing an ideal gas is heated while no gas enters or leaves the cylinder. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

194. The reaction $2\text{Ag}_2\text{O}(\text{s}) \longrightarrow 4\text{Ag}(\text{s}) + \text{O}_2(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 20. L. After 7.0 g of Ag_2O completely reacts while all other reactants are present in excess at $15.^\circ\text{C}$, determine the pressure of O_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

195. A gas mixture contains 7.2 g of fluorine (F_2) and 6.5 g of phosphine (PH_3). Determine the mole fraction of fluorine in the mixture. Justify your answer.

ANSWER: _____

196. A sample of $\text{C}_2\text{H}_5\text{OH}$ is transferred into a calibrated glass bulb having a measured volume of 11.1 L. After the gas reaches $11.^\circ\text{C}$, the pressure inside is measured to be 0.35 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

197. A gas is contained beneath a square piston with sides of 9.3-cm. The gas pressure is measured to be 9.1 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

198. A sample of 14.3 g of strontium carbonate reacts completely with excess hydrochloric acid to produce strontium chloride, water, and carbon dioxide gas in a sealed container. The container initially contains 0.19 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of carbon dioxide after the reaction is complete.

ANSWER: _____

199. A compressed-gas cylinder is connected to a large evacuated storage bag. The cylinder has an internal diameter of 7.52-cm, an internal height of 27-cm, and initially contains gas at 3.2 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

200. A fixed amount of gas is initially at 8.71°C . After its temperature is increased to 72°C and its volume becomes 22% smaller than its initial volume, its pressure is measured to be 12.0 atm. Determine the initial pressure. Report your answer in SI units and justify your answer.

ANSWER: _____